Neural Network

Friday, October 09, 2026 9:22 AM

#	A_1	A_2	X_3	Y
1	1	1	0	+
2	2	3	1	+
3	3	2	2	+
4	−8	−5	3	−
5	−7	−6	1	−

Perceptron find w = ?

$$\vec{X} = \begin{bmatrix} A_1 \\ A_2 \\ A_3 \\ 1 \end{bmatrix} \quad \vec{w} = \begin{bmatrix} w_1 \\ w_2 \\ w_3 \\ b \end{bmatrix} \qquad \eta = 0.2$$

initial $\vec{w}_0 = \begin{bmatrix} -1 \\ 0 \\ 1 \\ 1 \end{bmatrix}$ $\vec{X}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$

$\vec{X}_4 = \begin{bmatrix} 8 \\ 5 \\ -3 \\ -1 \end{bmatrix}$ $\vec{X}_5 = \begin{bmatrix} 7 \\ 6 \\ -1 \\ -1 \end{bmatrix}$

$\vec{w}_0 \cdot \vec{X}_1 = -1+0+0+1 = 0$ ✗ $\quad \vec{w}_1 = \eta(\vec{w}_0 + \vec{X}_1) = 0.2 \begin{bmatrix} 0 \\ 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.2 \\ 0.2 \\ 0.4 \end{bmatrix} = \vec{w}_1$

Step 1

$\vec{w}_1 \cdot \vec{X}_1 = 0 + 0.2 + 0 + 0.4 = 0.6$ ✓

$\vec{w}_1 \cdot \vec{X}_2 = 0 + 0.6 + 0.2 + 0.4 = 1.2$ ✓

$\vec{w}_1 \cdot \vec{X}_3 = 0 + 0.4 + 0.4 + 0.4 = 1.2$ ✓

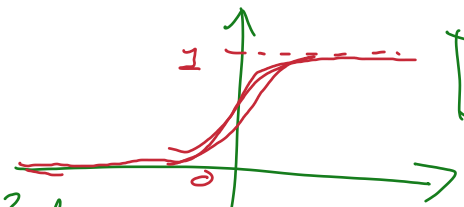
$\vec{w}_1 \cdot \vec{X}_4 = 0 + 1 - 0.6 - 0.4 = 0$ ✗

$$\vec{w}_2 = \eta(\vec{w}_1 + \vec{X}_4) = 0.2 \begin{bmatrix} 8 \\ 5.2 \\ -2.8 \\ -0.6 \end{bmatrix} = \begin{bmatrix} 1.6 \\ 1.04 \\ -0.56 \\ -0.12 \end{bmatrix}$$

✓

Sigmoid

$$\sigma(x) = \frac{1}{1+e^{-x}}$$



$$\sigma'(x) = \sigma(x)(1-\sigma(x))$$

$$\frac{d\sigma(x)}{dx} = \frac{d}{dx}(1+e^{-x})^{-1} = -1(1+e^{-x})^{-2} \frac{d(1+e^{-x})}{dx}$$

$$= \frac{-1}{(1+e^{-x})^2}(-e^{-x}) = \frac{e^{-x}}{(1+e^{-x})^2} = \frac{e^{-x}+1-1}{(1+e^{-x})^2} = \frac{e^{-x}+1}{(1+e^{-x})^2} 1 - \frac{1}{(1+e^{-x})^2}$$

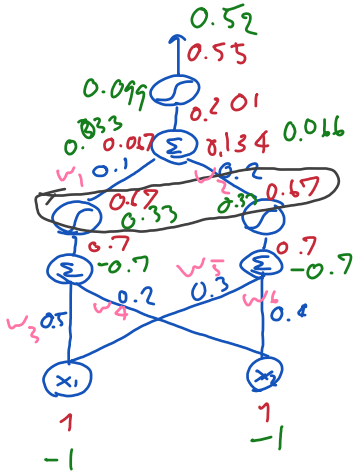
$$= \frac{1}{1+e^{-x}} - \frac{1}{(1+e^{-x})(1+e^{x})}$$

$$= \frac{1}{1+e^{-x}}\left(1 - \frac{1}{1+e^{x}}\right) = \sigma(x)(1-\sigma(x))$$

mini-batch = 1

$\eta = 0.1$

BP



ป้อน w 1 ชุด

	x_1	x_2	y	Output	t − o
①	1	1	1	0.55	0.45
②	−1	−1	0	0.52	−0.52

$\delta_k = o_k(1-o_k)(t_k - o_k)$

$\delta_1 = 0.55(1-0.55)(0.45) = 0.11$

$\delta_2 = 0.52(1-0.52)(-0.52) = -0.13$

$\delta = o(1-o)\sum w\delta$

hidden

① $O_1 = [0.67, 0.67]$ (0.1, 0.2)

② $O_2 = [0.33, 0.33]$ (0.1, 0.2)

$w' = w + \eta\delta x$

$$w = \underset{1\times 2}{[0.1\ \ 0.2]} + 0.1\underset{1\times 2}{[0.11\ \ -0.13]}\underset{2\times 2}{\begin{bmatrix}1 & 1\\ -1 & -1\end{bmatrix}}$$

$$= [0.1\ \ 0.2] + 0.1[0.24\ \ 0.24]$$

$$= [0.1\ \ 0.2] + [0.024\ \ 0.024]$$

$$= \underset{w_1}{[0.124}\ \ \underset{w_2}{0.224}]$$

PM Perceptron $\rightarrow [w, b] \rightarrow [w_1, w_2, w_3, b] = \vec{w}$

	a_1	a_2	a_3	target
①	1	1	0	1 → +1
②	2	2	1	1 → +1

step0 $y \cdot \vec{x}$

$\vec{x}_1 = [1\ \ 1\ \ 0\ \ 1]$

$\vec{x}_2 = [2\ \ 2\ \ 1\ \ 1]$

$\vec{x}_3 = [0\ \ 1\ \ 3\ \ -1]$

③ [0 | -1 | -3 | 1] -1

X Y

$\vec{w}_0 = [0, 0, 0, 0]$

$\vec{x}_1 \cdot \vec{w}_0 = 0$ ✗

$\vec{w}_1 = \vec{w}_0 + \vec{x}_1 = [1 \quad 1 \quad 0 \quad 1]$

$\vec{x}_1 \cdot \vec{w}_1 = 3$ ✓

$\vec{x}_2 \cdot \vec{w}_1 = 5$ ✓

$\vec{x}_3 \cdot \vec{w}_1 = 0$ ✗

$\vec{w}_2 = \vec{w}_1 + \vec{x}_3 = [1 \quad 2 \quad 3 \quad 0]$

w b

$w = 1 \quad 2 \quad 3$

$b = 0$

$a_1 + 2a_2 + 3a_3 + 0 = 0$

Sigmoid $\sigma(x) = \frac{1}{1+e^{-x}}$

$\frac{de^x}{dx} = e^x$

$\frac{d\sigma(x)}{dx} = \frac{d(1+e^{-x})^{-1}}{dx} = -1(1+e^{-x})^{-2} \frac{d(1+e^{-x})}{dx} = e^{-x}(1+e^{-x})^{-2}$

$= \frac{e^{-x}}{e^{+1-1}} = \frac{e^x}{e^x+1} - \frac{1}{(e^x+1)^2}$

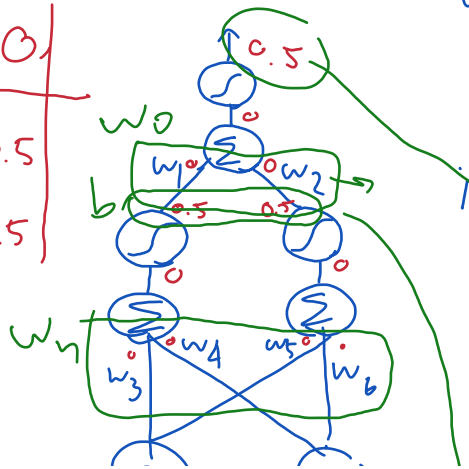
$$\frac{1}{(1+e^{-x})^2} \quad \frac{1}{(1+e^{-x})^2} \quad \frac{1}{(1+e^{-x})^2}$$

$$= \frac{1}{1+e^{-x}} - \frac{1}{(1+e^{-x})^2}$$

$$= \left(\frac{1}{1+e^{-x}}\right)\left[1 - \frac{1}{1+e^{-x}}\right] = \sigma(x)[1-$$

BP

	a_1	a_2	target	$\frac{1}{1+e^{-x}}$
x_1	1	−1	1	0.5
x_2	−1	1	0	0.5
	b_1	b_2		



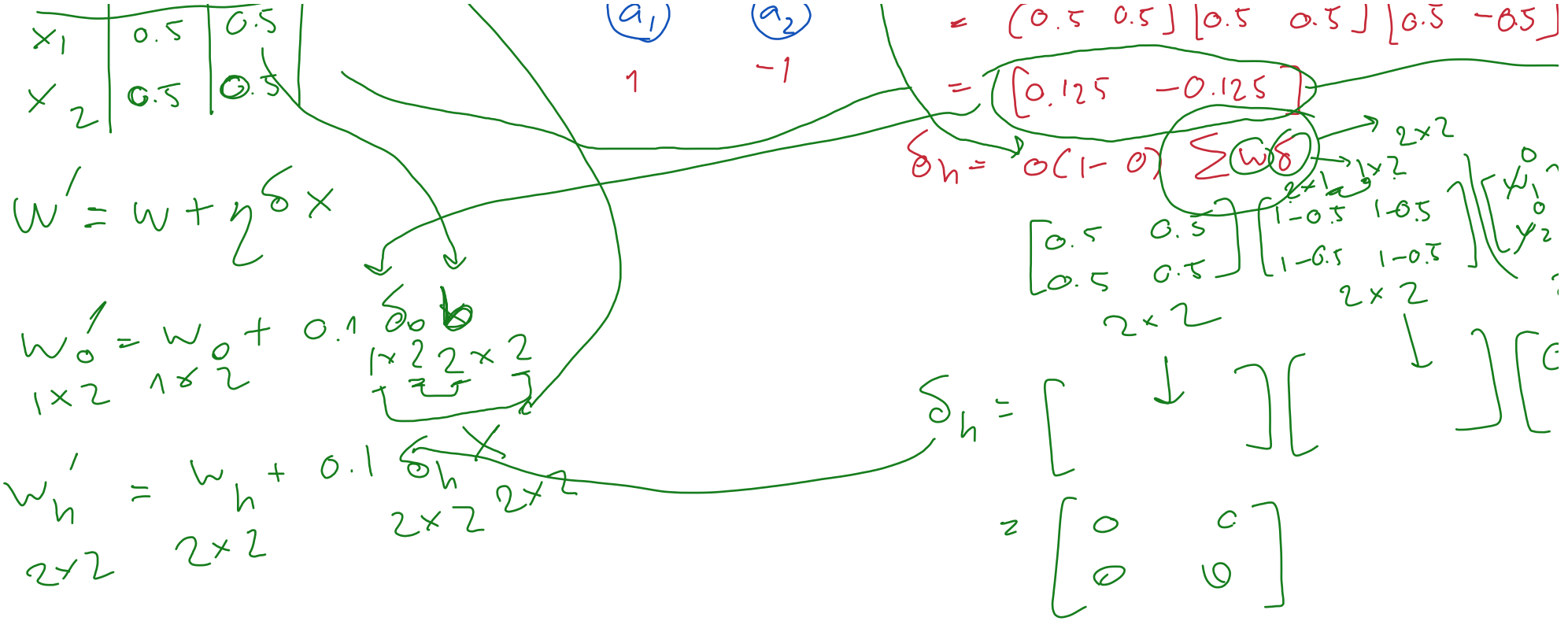
$w_1, w_2, w_3, w_4, w_5, w_6 = ?$

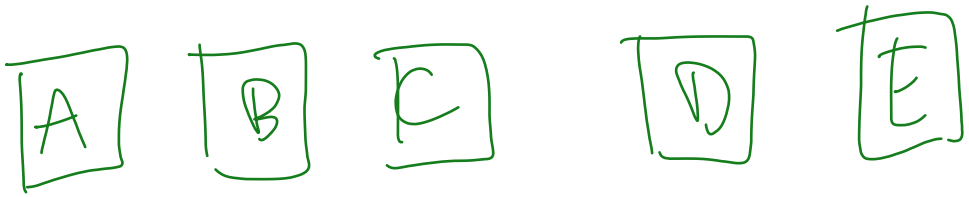
update 1 ครั้ง $\eta = 0.1$

init $w_0 = [w_1, w_2, 0, 0, 0, 0]$

$\delta_0 = o(1-o)(t-o)$

$= [0.5 \quad 0.5] \cdot [1-0.5 \quad 1-0.5] \cdot [1-o$


x_1 0.5 0.5
x_2 0.5 0.5
a_1 a_2
1 -1
$= [0.5\ 0.5]\ [0.5\ 0.5]\ [0.5\ -0.5]$
$= [0.125\ \ -0.125]$
$\delta_h = o(1-o)\sum w\delta$
$w' = w + \eta\delta x$
$w_o' = w_o + 0.1\,\delta_o\, b$
$1\times 2 \quad 1\times 2 \quad 2\times 2$
$w_h' = w_h + 0.1\,\delta_h\, x$
$2\times 2 \quad 2\times 2 \quad 2\times 2$
$\delta_h = [\quad][\quad]$
$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$



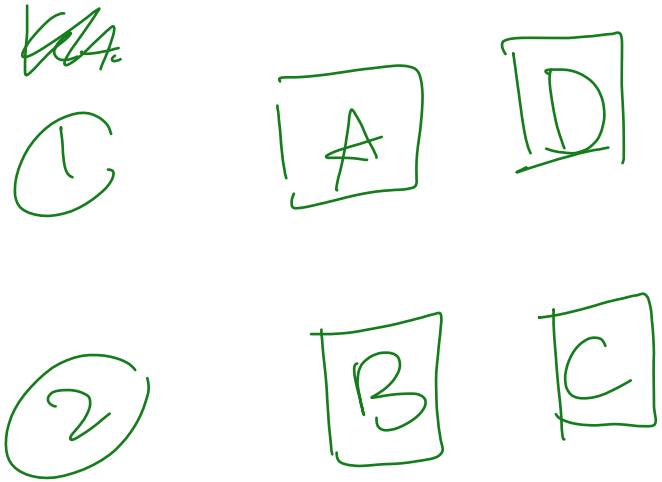
k = 2

k-means / FCM

k = ?

m = ?

k = ?



FC

input = #attr

hidden = free

output = #target

	age	gender	height	weight	fat or not
1	15	M	160	80	Y

input = 4
hidden = 8
output = 1

M, F → 0, 1

Dot

favor
V/C/S

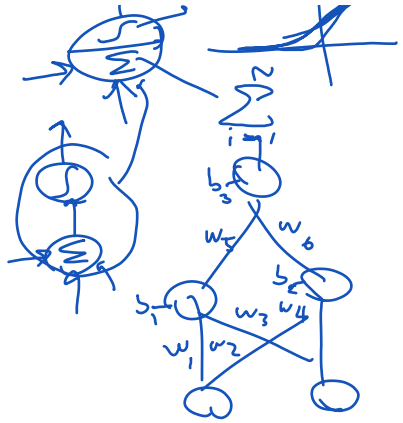
Pot = 2

	#leg	#arm	voice	Animal
1	4	0	bok	dog
2	4	0	meaw	cat
3	2	2	meaw	monkey

input = 3
hidden = 2
output = 3

		XOR

0	0	0
1	0	1
0	1	1
1	1	0



input = 2

output = 1

hidden = 2

9

2 - 2 - 1

#P = 9

$4 \underset{32}{-} 8 \underset{48}{-} 6 \underset{12}{-} 2$

#parameter = ~~16~~ ~~92~~

1 8+
32 6+
48 2
12
92 + 16 = 108

108 ✓

$10 \underset{20}{-} 2 \underset{10}{-} 5 \underset{30}{-} 6 \underset{48}{-} 8$

129 ✓

input dimension ≈ #attributes

output dimension = #class Y (target)

	age (attr X)	gender	weight	height	fat? (target)
1	15	M_0	80	160	Y
2	30 ✓	F_1	60 ✓	155 ✓	N
3					
4					

2 → 1
3 → 3
4 → 4

M = [0 1]
F = [1 0]

? [0 0 _ 0]
3

Degree of freedom

1 { C [1 0 0 0] N-1
L [0 1 0 0]
S [0 0 0 1]
V [0 0 1 0]

input = 4
hidden = 8, 3
ouput = 1

	favor (X)	leg	arm	animal (Y)
1	banana	2	2	monkey
2	papaya	4	0	hippo
3	carrot	4	0	rabbit
4	glass	4	0	ox
	3	1	1	

input = ? 5/6
output = ? 4

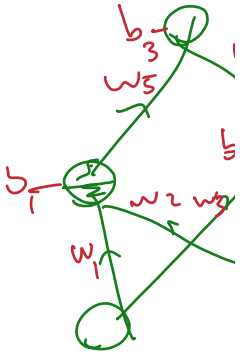
	x_1	x_2	bias	y
①	0	0	1	0
②	0	1	1	1
③	1	0	1	1
④	1	1	1	0

input = 2

output = 1

hidden = 2

$2(-2)+1 = 3$



$2 + 2 + 3 + 4 + 5 + 1$

$4 + 6 + 12 + 20 + 5 = 47 + 15 = 62$ #